GCI WORLD APRIL 2026: FINAL ASSIGNMENT

Churn Mitigating Strategies by leveraging Device and Equipment Demographics

A Business Proposal for Minimising Churning in a Telecom Company, through Data Analysis

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CURRENT MARKET ANALYSIS OF TELECOMMUNICATIONS INDUSTRY

1. Monthly churn rate figures:
   - Churn rates in well known telecom companies in the USA have been reported as follows: AT&T ~0.98-1.12% monthly, T-Mobile ~0.90-1.02% monthly, Verizon ~1.1%+ in late 2025.
   - Industry average roughly **under 1% monthly** for postpaid phone churn, translating to **roughly 10-12% annualized** churn typically cited in industry reports, forcing major carriers to replace millions of subscribers yearly just to sustain scale.

2. The High Cost of Acquisition (CAC):
   - Acquiring a new customer is widely cited as **5x to 10x more expensive** (ranging from 3x to 25x) than retaining an existing one.

3. Smartphone replacement cycle and refurbished phones.
   - Globally, people replace their smartphone **every 2.4-3.5 years**. The shortest replacement cycle is seen in North America, where smartphones are replaced **every 20 months**.
   - Secondary smartphone markets are growing rapidly, making up roughly **17% of total global shipments**.
   - Certified refurbished/pre-owned options cater to budget segments, offering **30%–40% cost savings** and thus hurting new smartphone sales.

^Multiple references have been used for the market analysis. See list of references for details.

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INTRODUCTION OF DATA

Historical Data of Company A (an anonymous telecommunications service provider), containing customer information and usage history of approximately 100,000 clients.

record.csv

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	rev_Mean	mou_Mea	totmrc_M	da_Mean	ovrmou_N	ovrrev_M	vceovr_M	datovr_M	roam_Me	change_m	change_re	drop_vce_	drop_dat_	blck_vce_l	blck_dat_l	unan_vce_	unan_dat_	plcd_vce_
2	23.9975	219.25	22.5	0.2475	0	0	0	0	0	-157.25	-18.9975	0.666667	0	0.666667	0	6.333333	0	52.33333
3	57.4925	482.75	37.425	0.2475														
4	16.99	10.25	16.99	0														
5	38	7.5	38	0														
6	55.23	570.5	71.98	0														
7	82.275	1312.25	75	1.2375														
8	17.145	0	16.99	0														
9	38.0525	682.5	52.49	0.2475														
10	97.3375	1039	50	4.95														

client.csv

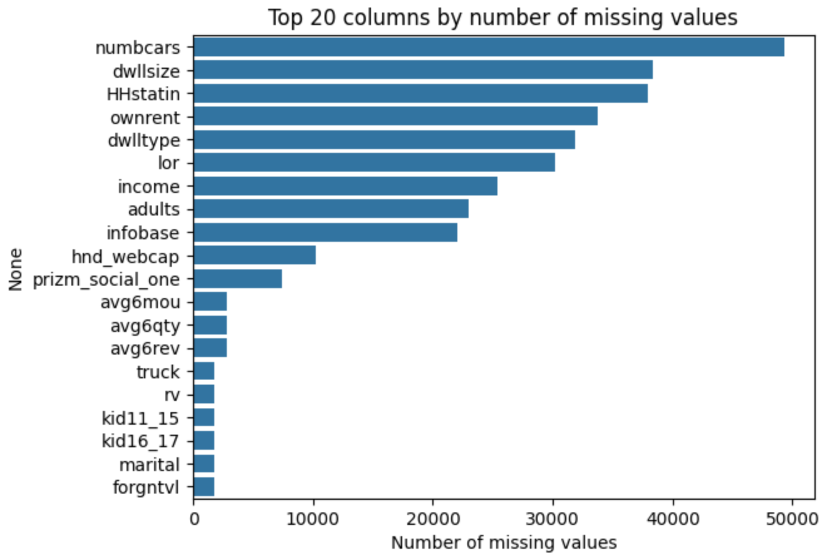
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	uniqsubs	actvsubs	new_cell	crclscod	asl_flag	totcalls	totmou	totrev	adjrev	adjmou	adjqty	avgrev	avgmou	avgqty	avg3mou	avg3qty	avg3rev	avg6mou	avg6qty
2	2	1	U	A	N	1652	4228	1504.62	1453.44	4085	1602	29.66	83.37	32.69	272	116	30	322	136
3	1	1	N	EA	N	14654	26400	2851.68	2833.88	26367	14624	51.53	479.4	265.89	305	158	40	477	275
4	1	1	Y	C	N	7903	24385.05	2155.91	1934.47	24303.05	7888	34.54	433.98	140.86	12	7	17	11	6
5	1	1	Y	B	N	1502	3065	2000.9	1941.81	3035	1479	40.45	63.23	30.81	8	3	38	50	25
6	1	1	Y	A	N	4485	14028	2181.12	2166.48	13965	4452	38.69	249.38	79.5	558	191	55	586	196
7	2	2	Y	C	N	26812	40869	4033.98	3932.9	40295	26362	83.68	857.34	560.89	1260	960	80	1187	853
8	2	2	N	A	N	6279	17390.03	3091.7	3065.24	17371.03	6271	58.95	334.06	120.6	0	0	17	0	0
9	1	1	Y	B	N	4491	12492	1427.71	1423.06	12439	4470	34.71	303.39	109.02	633	96	39	719	112
10	1	1	Y	B	N	16730	43231.05	4404.44	4313.71	42943.05	16579	81.39	810.25	312.81	973	465	90	915	434

Data Contents:

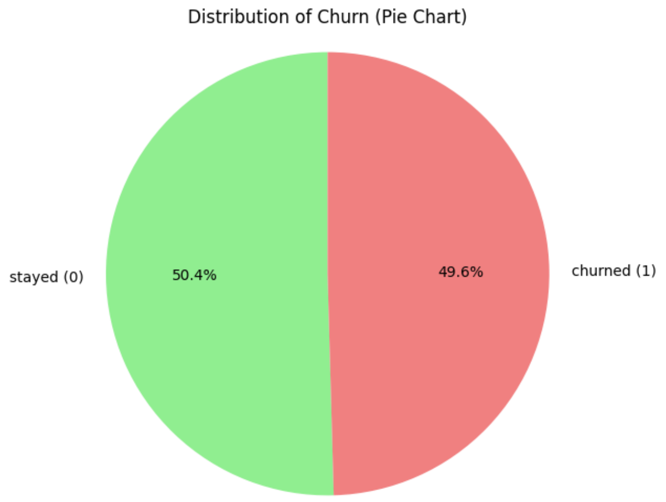
- Key data for each customer (demographics, family information, and behavioral flags)
- Information on customers’ credit classification and account limits
- Monthly billing and revenue status of customers
- Monthly voice and data usage information (minutes and call types)
- Call connectivity Quality of Service (QoS) metrics
- -Total lifetime account activity and aggregated historical volumes
- Rolling historical averages (3-month and 6-month horizons)

DATA VISUALISATION

We conducted various visualizations of the customer data.

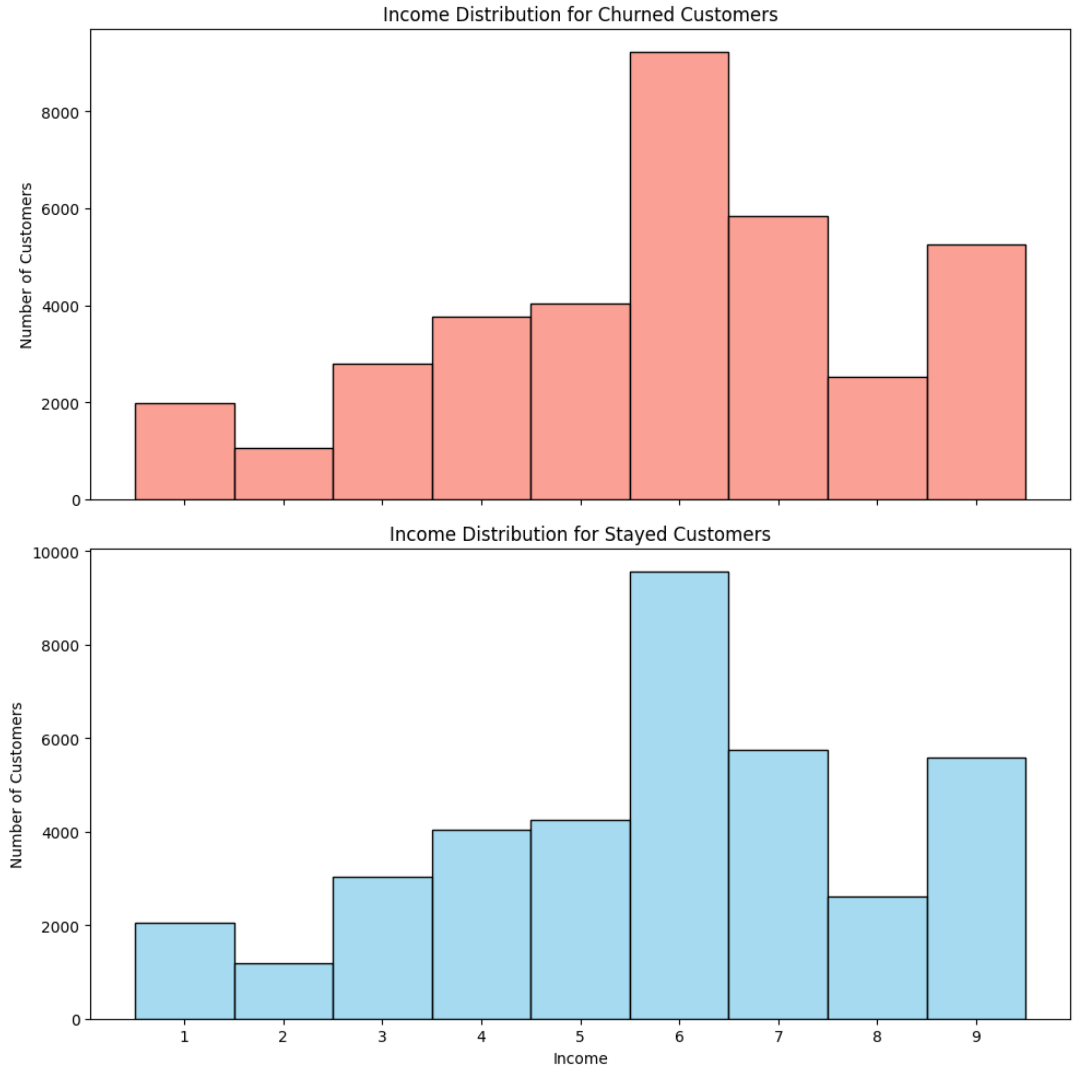


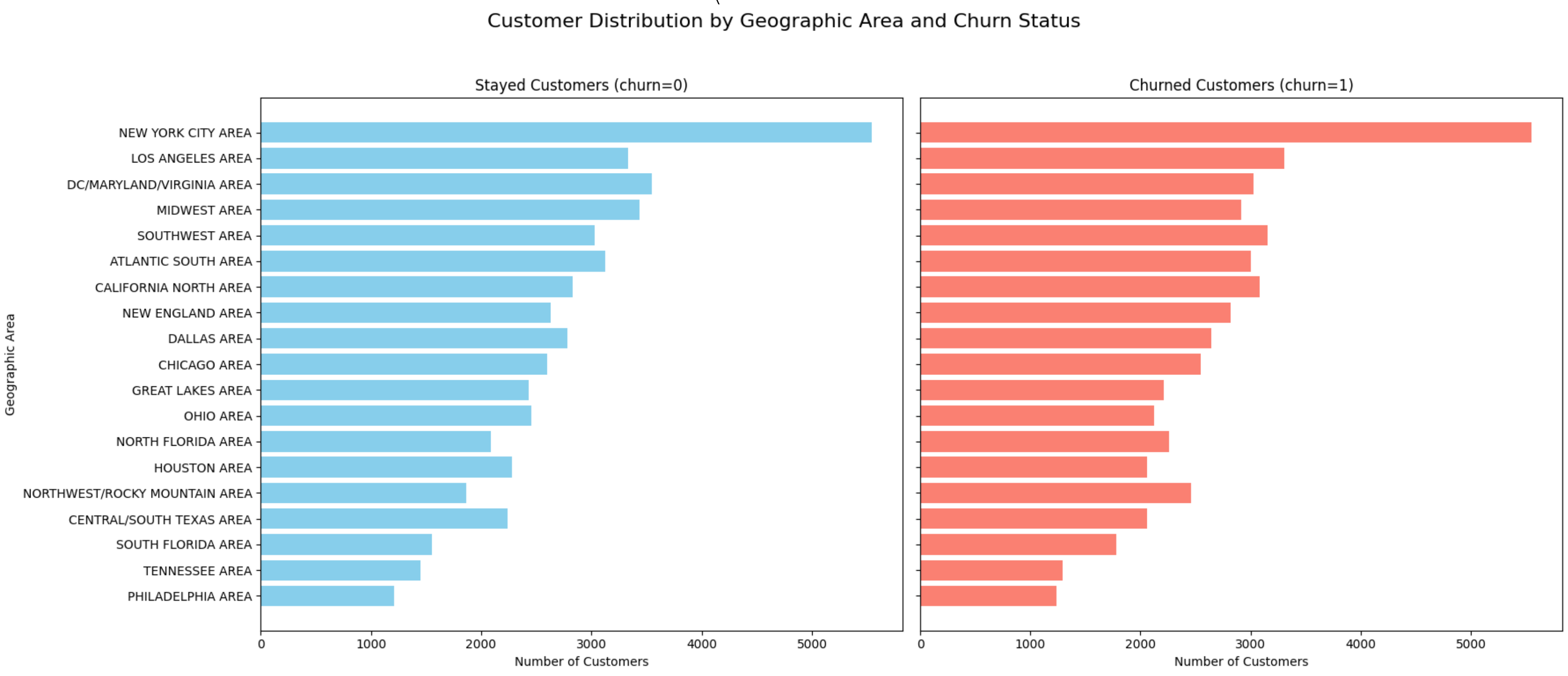
Columns with mostly missing values provide little info for our analysis and dropping them is the safest option



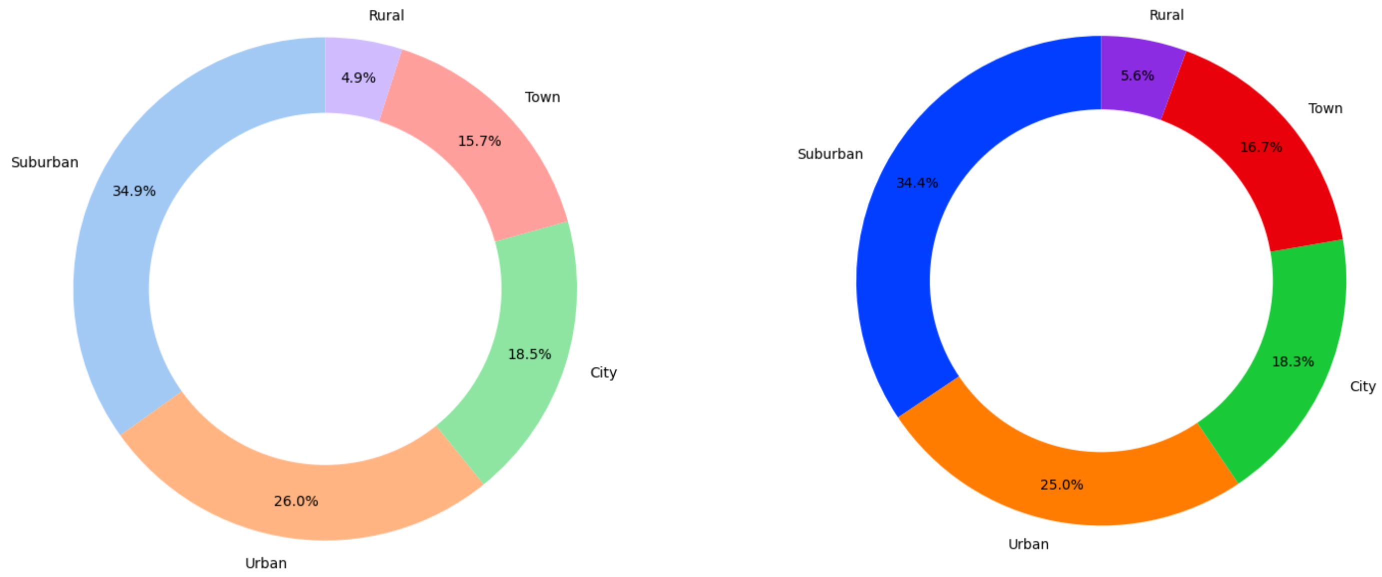
Our **target variable** is 'churn' and from pie chart, we can see it is a balanced feature.

Income distributions for stayed and churned clients





Geographical area statistics for stayed and churned clients



It is difficult and cumbersome to distinguish staying customers from churn customers by analysing individual features one by one.

Advanced analysis using machine learning methods is necessary, to find appropriate features which affect churn.

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OVERVIEW OF MACINE LEARNING MODEL

To find features which have high importance in predicting a certain target variable (which here is churn), we will use XGBoost.

Input

X → Target Variable, which we want to predict
Y → All the other columns/features in dataset, except the target variable

Dataset Splitting

70/30 split for training and testing; using train_test_split

Model Definition

```
model = XGBClassifier(
n_estimators=300,
learning_rate=0.05,
max_depth=6,
eval_metric='logloss',
random_state=42, )
```

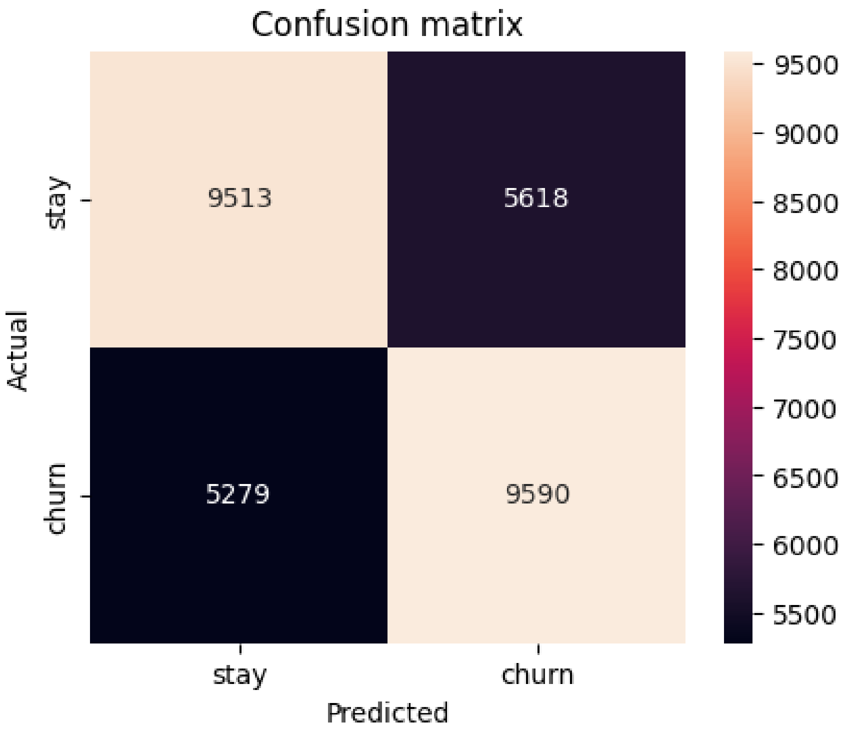
Training

Model is trained, internally building a network of 300 trees between features and target variable.

Prediction

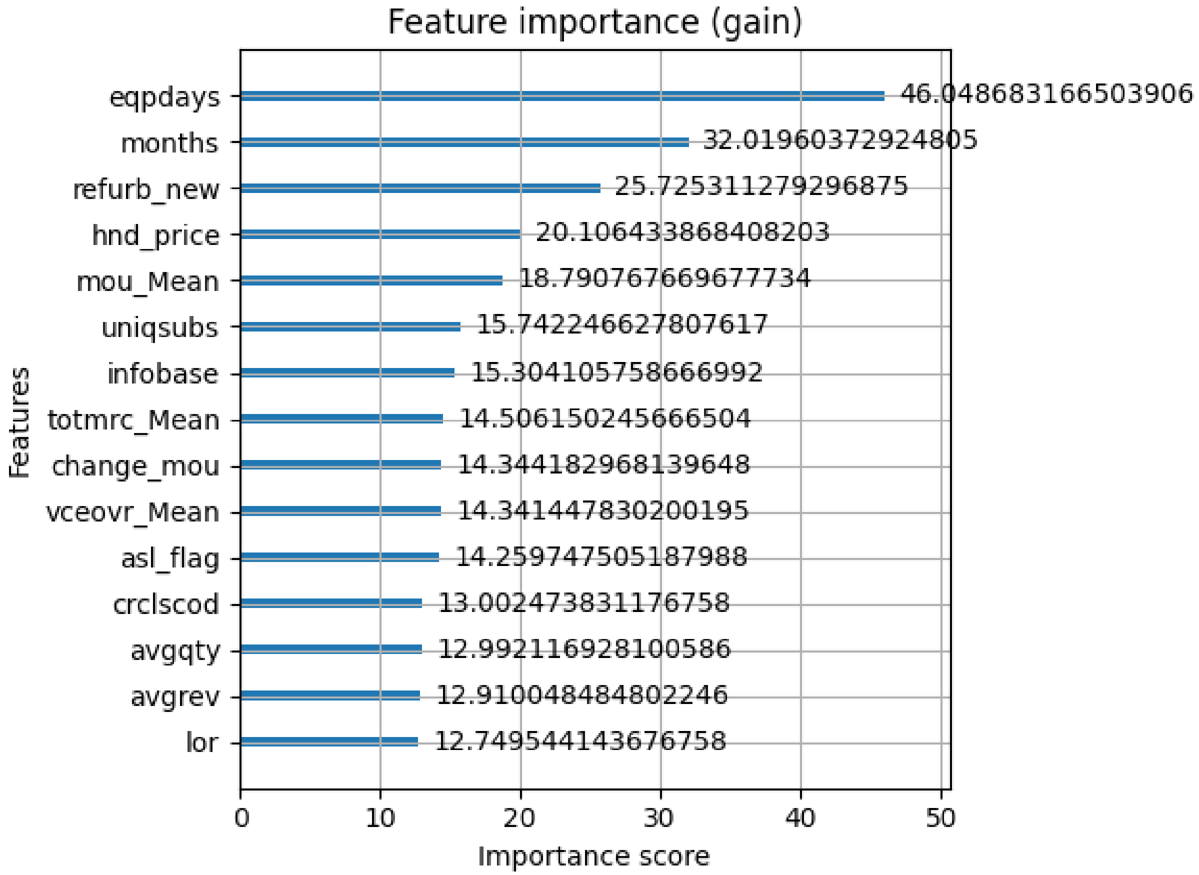
Outputs predictions (y_pred) and probability (y_proba)

We run the XGBoost model on churn, our target variable:



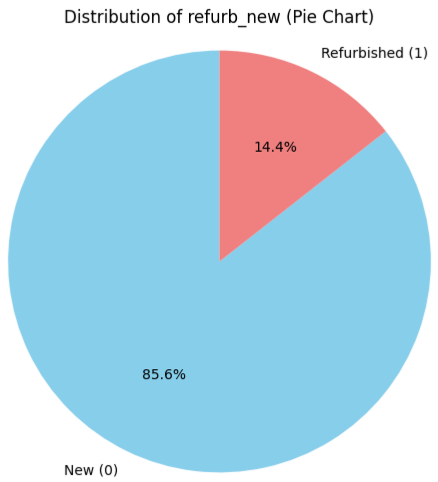
Predictions were plotted in a confusion matrix.

Accuracy was calculated to be 0.6368. This is a useful metric, as churn is a balanced class



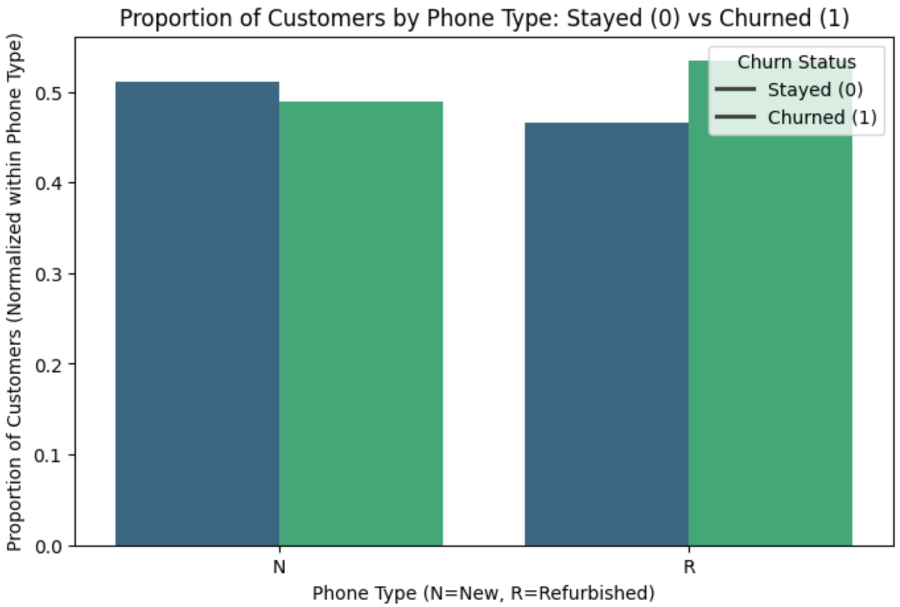
Features with highest gain in XGBoost are plotted, which gives a measure of how important the feature is in predicting the target variable

We will be basing our proposal on refurb_new (Clients with refurbished vs new handsets). It is among the top 5 features with high importance in predicting churn.

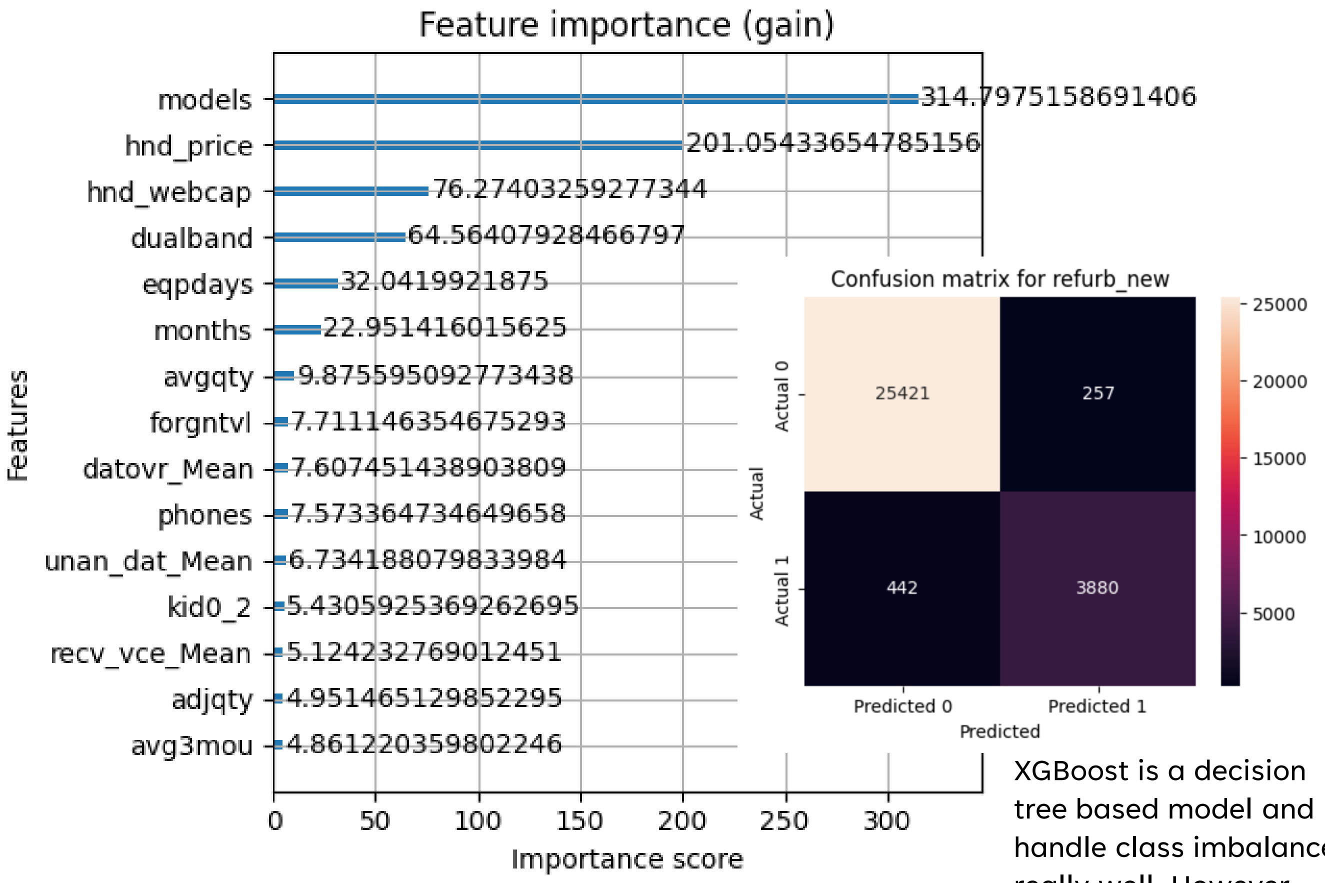


refurb_new is an imbalanced feature

What features can predict refurb_new? We run the XGBoost model again with target variable as refurb_new.

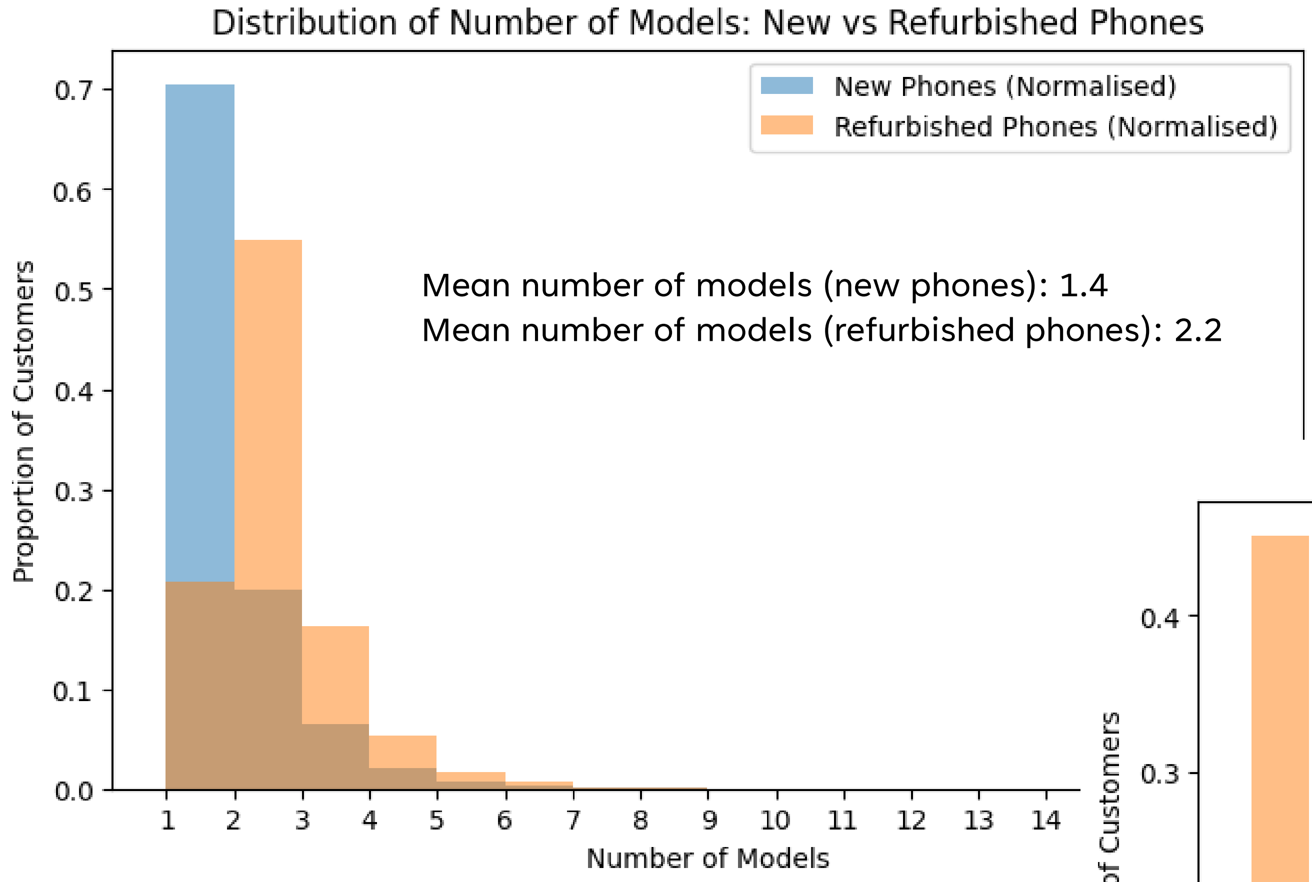


The above bar graph is normalised, to combat class imbalance. We can see that there are more instances of churn than stay among customers with refurbished handsets.

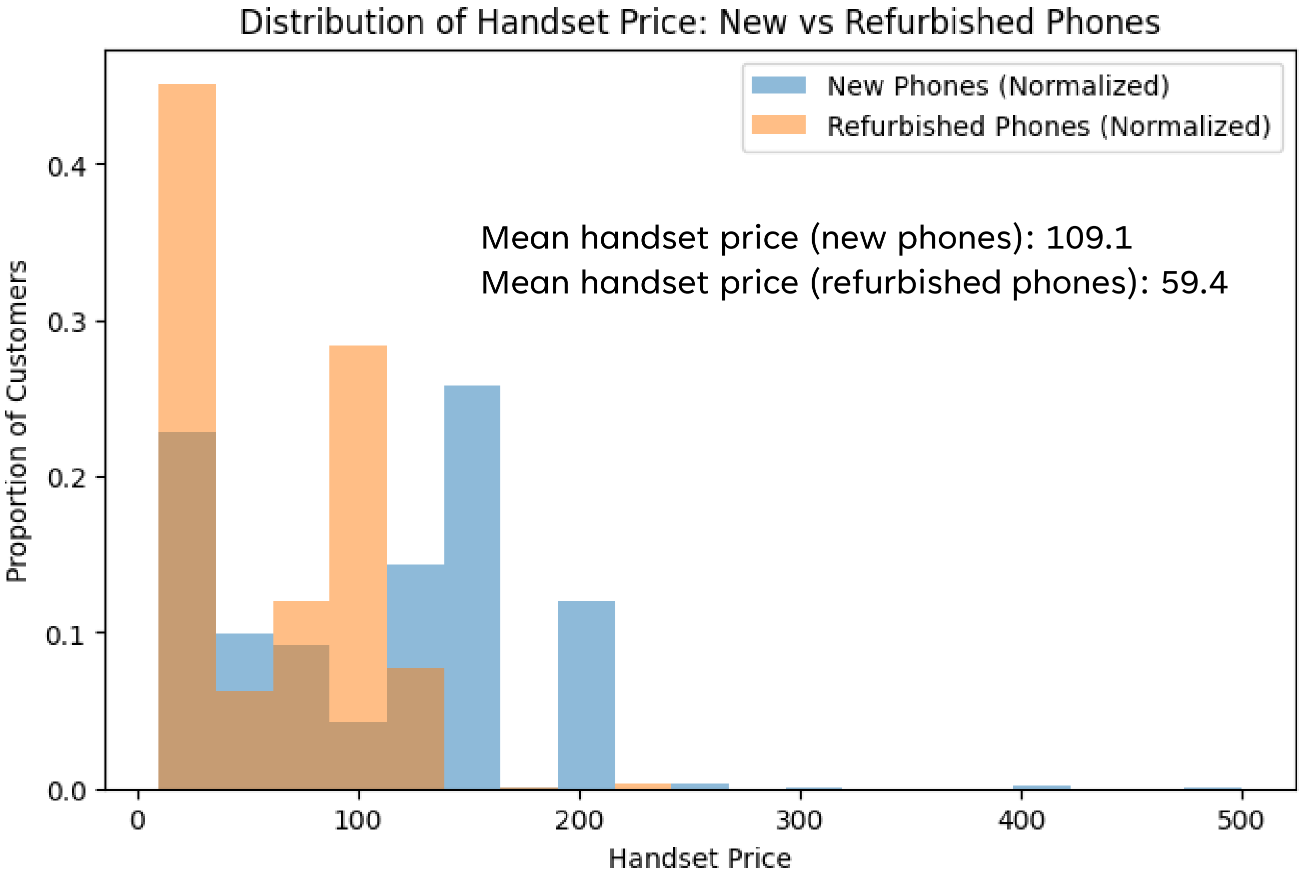


models (no. of models issued) and hnd_price (handset price) seem to be good features to relate to refurb_new.

XGBoost is a decision tree based model and handle class imbalance really well. However, accuracy is no longer a meaningful metric. F1-score is a better metric for such an instance, and it is reported to be 0.9174.



No. of models issued is typically higher in in customers with refurbished handsets, as mean and right tail are both higher for refurbished handsets than for new handsets.



Price of handset is typically higher for new handsets, as mean and right tail are both higher for new handsets than for refurbished handsets.

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INFERENCES

- We have seen that churn depends on refurb_new (refurbished vs new handset) from the XGBoost model.
- However, we cannot simply make customers transition to new handsets from refurbished handsets.
- That is why, we used XGBoost to find other features that we can relate to refurb_new, which we found to be models (no. of models issued) and hnd_price (price of handset). This gave us two important results:
  - ➢ Customers with refurbished phones typically have more number of models issued, which suggests that such customers change their phones frequently, and this maybe because the handset is not fulfilling their needs enough.
  - ➢ Refurbished handsets tend to be much less costlier than new handsets, which might be the reason why customers with refurbished handsets gravitate towards refurbished handsets: to be cost effective.

These are the levers that we can use.

BUSINESS PROPOSAL

- We will target customers with refurbished phones, who have greater number of models issued. These are customers who may churn. We can also identify the highest revenue customers among these customers, and they will be the highest churn risk customers i.e. if these customers churn, it will be highly detrimental to the company revenue, and we must act upon these customers first.
- We will approach these high risk customers and offer the following
  - ➢ A better mobile plan than their current plan, with more mobile data, incoming/outgoing calls, text messages etc.
  - ➢ Other local benefits, like free access to streaming platforms for a certain duration, perks etc.
  - ➢ And finally, the most important, a high discount (or maybe even a complementary) upgrade to a new handset, with mandatory service renewal of 12-24 months, hence locking down revenue for that period.
- All these discounts may seem detrimental to the company, but in the long run it is actually better for the company, here is how:
  - ➢ Due to the multiple benefits, there is significantly high chance that the customer will stay rather than churn and thus, the customer's total lifetime revenue for the company will definitely increase.
  - ➢ From multiple references, we can see that customer acquisition costs are significantly higher than customer retention costs^. By providing these benefits to customer and making him stay, we are saving money for the company by paying only customer retention costs rather than new customer acquisition costs to replace the churned customer.

  Therefore, customer lifetime value and saving on customer acquisition costs outweighs the losses due to discounts and increased customer benefits.

* Even though, in the given dataset, number of customers with refurbished handsets is low compared to those with new handsets, it is still a major feature to predict churn. If in other companies, customers with refurbished phones are more than 50%, implementing such a proposal will be a must to not lose business!

^ churnkey.co/blog (See list of references).